Grammar :

Lecture-3
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Four important components of Grammar

1. Terminals, or terminal symbols (T)
2. Non-terminals or syntactic categories (V)
3. start symbol (S)
4. Production or rules (P)

Grammar, $G = (V, T, P, S)$

V : Variable

Grammar for Palindroms

$S \longrightarrow P$

$P \longrightarrow 0P0 \mid 1P1$

$P \longrightarrow 0 \mid 1 \mid \epsilon$

$\therefore G_{pal} = (\{P\}, \{0,1\}, P, S)$

P : production, {0,1} : terminal, S : start, {P} : variable

Derivation Using Grammar

Derivation:

The language of the grammar is all strings of terminals that we can obtain in head to body manner. This use of grammars is called derivation.

Example:

$S \longrightarrow P$

$\longrightarrow 0P0$

$\longrightarrow 01P10$

$\longrightarrow 01\epsilon10 \longrightarrow 0110$

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Left most derivation :

In a derivation, if we replace the leftmost variable by one of its production bodies, such derivation is called a leftmost derivation.

step by step

Example:

$S \longrightarrow E$

$E \longrightarrow E+E \mid E-E$

$E \longrightarrow I$

$I \longrightarrow id$

string: id-id+id

Left most derivation:

$$S \xrightarrow{lm} E$$
$$\xrightarrow{lm} E-E$$
$$\xrightarrow{lm} I-E$$
$$\xrightarrow{lm} id-E$$
$$\xrightarrow{lm} id-E+E$$
$$\xrightarrow{lm} id-I+E$$
$$\xrightarrow{lm} id-id+E$$
$$\xrightarrow{lm} id-id+I$$
$$\xrightarrow{lm} id-id+id$$

Right most derivation :

In a derivation, if we replace the rightmost variable by one of its production bodies step by step, such derivation is called a rightmost derivation.

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